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United States Patent	12385630
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Colton; Mark et al.

Portable flashlight

Abstract

A portable flashlight apparatus is provided that is lightweight and compact to provide enhanced illumination in an ambient environment when placed in one of a plurality of operating modes. The portable flashlight apparatus includes a base housing defining an interior space and which includes structurally defined regions that include a front housing region, a rear housing region, an intermediate housing region extending between the front housing region and the rear housing region and an end cap that closes the interior space at the rear housing region. A light module includes a first light source operable to emit illuminating light, a second light source operable to emit laser light, and a third light source operable to emit illuminating light. A magnet are arranged at the rear face of the end cap, and is operable to magnetically bias the base housing on a working surface or support surface having magnetic properties.

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Appl. No.: 18/314520

Filed: May 09, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230383932 A1	Nov. 30, 2023

Related U.S. Application Data

Publication Classification

Int. Cl.: **F21V21/096** (20060101); **F21L4/02** (20060101); **F21L4/08** (20060101); **F21V21/088** (20060101); **F21Y115/10** (20160101); F21Y115/30 (20160101)

U.S. Cl.:

CPC **F21V21/0965** (20130101); **F21L4/025** (20130101); **F21L4/027** (20130101); **F21L4/085** (20130101); **F21V21/0885** (20130101); F21Y2115/10 (20160801); F21Y2115/30 (20160801)

Field of Classification Search

CPC: F21V (21/0965); F21V (21/0885); F21V (23/0428); F21V (23/06); F21L (4/005); F21L (4/085); F21L (4/022); F21L (4/027); F21Y (2115/10); F21Y (2113/00); F21Y (2115/30)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to and the benefit of U.S. Provisional Application No. 63/345,291 filed on May 24, 2022, entitled PORTABLE FLASHLIGHT, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(1) One or more embodiments of the present disclosure relate generally to a portable flashlight apparatus that provides illumination in an ambient environment when placed in one of a plurality of operating modes.

BACKGROUND

(2) Portable flashlights are used to provide selective illumination in relatively small access spaces and/or areas.

SUMMARY

(3) In accordance with one or more example embodiments, a portable flashlight apparatus is provided that is lightweight and compact to provide enhanced illumination in an ambient environment when placed in one of a plurality of operating modes. The portable flashlight apparatus includes a base housing and a light module.

(4) In accordance with one or more example embodiments, the base housing includes a plurality of structurally-defined regions that include a front housing region at a front end of the portable flashlight apparatus, a rear housing region at a rear end of the portable flashlight apparatus, and an intermediate housing region extending between the front housing region and the rear housing region. The base housing defines an interior space or opening to receive a holder upon which is mounted the light module that supports a plurality of light sources operable to illuminate light, and a rechargeable power source for providing power to the light sources.

(5) In accordance with one or more example embodiments, one or more seal members, including a first seal member arranged between the rear housing region and an end cap, and a second seal member arranged at the front housing region, are provided to prevent moisture, debris, particles, etc. from entering into the interior space of the base housing and adversely affecting the performance of the light module and relevant components arranged in the interior space.

(6) In accordance with one or more example embodiments, the outer perimeter of the intermediate housing region defines a gripping surface that permits a user to maintain a grip of the base housing. The front housing region and the rear housing region respectively have an outer perimeter that is greater than the outer perimeter of the intermediate housing region, thereby permitting a user to maintain an enhanced grip of the base housing during use and/or transport of the portable flashlight apparatus.

(7) In accordance with one or more example embodiments, the portable flashlight apparatus includes a clip member that extends longitudinally along outer sidewall surfaces of the rear housing region and the intermediate housing region to facilitate secure attachment of the portable flashlight apparatus to a user. A hook member is arranged on a rear face of the end cap and is pivotably moveable between a stowed position on the rear face of the end cap and a deployed position in a direction away from the rear face of the end cap to facilitate hanging or suspension of the portable flashlight apparatus at a support point.

(8) In accordance with one or more example embodiments, a magnetic bias is provided to anchor or maintain the portable flashlight apparatus on a working surface or support surface having magnetic properties. One or more magnetic elements are integrated in, arranged on, or embedded in the rear face of the end cap to facilitate a magnetic attraction between the base housing and the working surface or support surface that magnetically biases the base housing on the working surface or support surface. In that way, hands-free use of the portable flashlight apparatus by a user is facilitated.

(9) In accordance with one or more example embodiments, the light module is arranged at the front housing region, and comprises a plurality of light sources. The light sources include a first light source arranged at a front face of the front housing region. The first light source comprises one or more LEDs operable to emit illuminating light (e.g., in a spotlight operating mode) from the front housing region in a direction parallel to the longitudinal axis of the base housing. A second light

source is arranged at a front face of the front housing region **201**. The second light source comprises a laser operable to emit laser light from the front housing region in a direction parallel to the longitudinal axis of the base housing. A third light source is arranged at a sidewall of the front housing region. The third light source comprises one or more LEDs operable to emit illuminating light (e.g., in a floodlight operating mode) from the front housing region in a direction perpendicular to the longitudinal axis of the base housing.

(10) In accordance with one or more example embodiments, the color temperature of light emitted by the LEDs of the first light source and the third light source is 4,500 K, respectively.

(11) In accordance with one or more example embodiments, the rechargeable power source is arranged at the intermediate housing region. A power charging port is located on an outer sidewall surface of the intermediate housing region to facilitate recharging of the rechargeable power source. The power charging port comprises a USB-C (also known as USB Type-C) port that facilitates a wired connection with a male plug of a USB-C charging cable for recharging of the rechargeable power source. The USB-C port is operable to facilitate a total charging time of 60 minutes to reach a full charged state of the rechargeable power source.

(12) In accordance with one or more example embodiments, a charge indicator is located on the outer sidewall surface of the intermediate housing region to visually indicate a charge status of the rechargeable power source during a recharging sequence. The charge indicator is operable to illuminate a single color that corresponds to the charge status of the rechargeable power source. Alternatively, the charge indicator is operable to illuminate a plurality of colors that correspond to the charge status of the rechargeable power source.

(13) In accordance with one or more example embodiments, the portable flashlight apparatus includes a user interface (UI) comprising a first control switch member and a second control switch member arranged adjacent to the first control switch member on the outer sidewall surface of the front housing region.

(14) In accordance with one or more example embodiments, the first control switch member is operable to control activation/deactivation of the LEDs of the first light source and the third light source, respectively between a plurality of operating modes. The second control switch member is operable to control activation/deactivation of the laser light of the second light source between the plurality of operating modes.

(15) In accordance with one or more example embodiments, the one or more LEDs of the first light source and the third light source are operable for manual control by a user via the first control switch member to emit illuminating light. The first control switch member is also operable to control an output/intensity of light emitted by the first light source and the third light source between the plurality of operating modes. For example, in a first or maximum (high) spotlight operating mode of the first light source, an output of the first light source is 1200 lumens with a runtime of 4 hours. In a second or maximum (low) spotlight operating mode, an output of the first light source is 1000 lumens with a runtime of 2 hours. In a third or minimum (high) spotlight operating mode, an output of the first light source is 200 lumens with a runtime of 8 hours. In a fourth or minimum (low) spotlight operating mode, an output of the first light source is 150 lumens with a runtime of 6 hours. In a fifth or maximum floodlight operating mode, an output of the third light source is 250 lumens with a runtime of 6 hours. In a sixth or minimum floodlight operating mode, an output of the third light source is 150 lumens with a runtime of 4 hours.

Description

DRAWINGS

(1) The various advantages of one or more exemplary embodiments will become apparent to one skilled in the art by reading the following specification and appended claims, and by referencing

the following drawings, in which:

(2) FIG. 1 illustrates a top, perspective view of an example portable flashlight apparatus, in accordance with one or more embodiments set forth, shown, and described herein.

(3) FIG. 2 illustrates a bottom, perspective view of the example portable flashlight apparatus of FIG. 1.

(4) FIG. 3 illustrates a front view of the example portable flashlight apparatus of FIG. 1.

(5) FIG. 4 illustrates a rear view of the example portable flashlight apparatus of FIG. 1.

(6) FIG. 5 illustrates a side view of the example portable flashlight apparatus of FIG. 1.

(7) FIG. 6 illustrates a side view of the example portable flashlight apparatus of FIG. 1.

(8) FIG. 7 illustrates a top view of the example portable flashlight apparatus of FIG. 1.

(9) FIG. 8 illustrates a bottom view of the example portable flashlight apparatus of FIG. 1.

(10) FIG. 9 illustrates an exploded view of the example portable flashlight apparatus of FIG. 1.

DESCRIPTION

(11) Turning to the figures, in which FIGS. 1-9 illustrates an example portable flashlight apparatus **100**, in accordance with one or more embodiments. The portable flashlight apparatus **100** may comprise one or more operational elements. Some of the possible operational elements of the portable flashlight apparatus **100** are illustrated in FIG. 1 and will now be described. It will be understood that it is not necessary for the portable flashlight apparatus **100** to incorporate all the elements illustrated in the figures set forth, and/or described herein. The portable flashlight apparatus **100** may have any combination of the various elements illustrated in FIG. 1. Moreover, the portable flashlight apparatus **100** may have additional operational elements to those illustrated in FIG. 1.

(12) As illustrated in FIG. 1, the example portable flashlight apparatus **100** comprises a base housing **200** and a light module **300**.

(13) Base Housing

(14) As illustrated in FIGS. 1 and 2, the base housing **200** includes an elongated body having a plurality of structurally-defined regions that include a front housing region **201** at a front end of the portable flashlight apparatus **100**, a rear housing region **202** at a rear end of the portable flashlight apparatus **100**, and an intermediate housing region **203** extending between the front housing region **201** and the rear housing region **202**.

(15) As illustrated in FIG. 9, the base housing **200** defines an interior space or opening to receive a holder **208** upon which is mounted the light module **300** which supports a plurality of light sources **301**, **302**, **303** operable to illuminate light, and a rechargeable power source **400** for providing power to the light sources.

(16) A first seal member **209** is arranged between the rear housing region **202** and an end cap **204**, and a second seal member **210** is arranged at the front housing region **201** to prevent moisture, debris, particles, etc. from entering into the interior space of the base housing **200** and adversely affecting the performance of the light module **300** and relevant components arranged in the interior space.

(17) The base housing **200** may be composed in whole or in part of a lightweight metal such as, for example, aluminum. Embodiments, however, are not limited thereto, and thus, this disclosure contemplates the base housing **200** being composed of any suitable material that falls within the spirit and scope of the principles of this disclosure. Although the illustrated example embodiments feature the base housing **200** having a generally rectangular cross-section, embodiments are not limited thereto. This disclosure contemplates the base housing **200** having any geometric cross-section that will fall within the spirit and scope of the principles of this disclosure.

(18) The outer perimeter of the intermediate housing region **203** defines a gripping surface that permits a user to maintain a grip of the base housing **200**. The front housing region **201** and the rear housing region **202** respectively have an outer perimeter that is greater than the outer perimeter of the intermediate housing region **203** to permit a user to maintain an enhanced grip of the base

housing **200** during use and/or transport of the portable flashlight apparatus **100**.

(19) As illustrated in FIGS. **3**, **4** and **9**, a clip member **205** extends longitudinally along outer sidewall surfaces of the rear housing region **202** and the intermediate housing region **203** to facilitate secure attachment of the portable flashlight apparatus **100** to a user. A hook member **206** is arranged on a rear face of the end cap **204** and is pivotably moveable between a stowed position on the rear face of the end cap **204** and a deployed position in a direction away from the rear face of the end cap **204** in order to facilitate hanging or suspension of the portable flashlight apparatus **100** at a support point.

(20) A magnetic bias is provided to anchor or maintain the portable flashlight apparatus **100** on a working surface or support surface having magnetic properties. One or more magnetic elements **207** are integrated in, arranged on, or embedded in the rear face of the end cap **204**. The one or more magnetic elements **207** facilitate a magnetic attraction between the base housing **200** and the working surface or support surface that magnetically biases the base housing **200** on the working surface or support surface. In that way, hands-free use of the portable flashlight apparatus **100** is facilitated.

(21) Light Module

(22) As illustrated in FIGS. **3** and **4**, the light module **300** is arranged at the front housing region **201** and comprises a plurality of light sources. The light sources include a first light source **301** arranged at a front face of the front housing region **201**, and which comprises one or more LEDs operable to emit illuminating light (e.g., in a spotlight operating mode) from the front housing region **201** in a direction parallel to the longitudinal axis of the base housing **200**. A second light source **302** is arranged at a front face of the front housing region **201** and comprises a laser operable to emit laser light from the front housing region **201** in a direction parallel to the longitudinal axis of the base housing **200**. A third light source **303** is arranged at a sidewall of the front housing region **201** and comprises one or more LEDs operable to emit illuminating light (e.g., in a floodlight operating mode) from the front housing region **201** in a direction perpendicular to the longitudinal axis of the base housing **200**. The one or more LEDs of the first light source **301** are arranged in an array on a printed circuit board (PCB). During activation of the first light source **301**, heat generated by the LEDs is transmitted to a heat sink, which radiates the heat away from the PCB. The one or more LEDs of the third light source **303** are arranged in an array on a second PCB. During activation of the second light source **303**, heat generated by the LEDs is transmitted to the heat sink, which radiates the heat away from the second PCB.

(23) In accordance with one or more example embodiments, the color temperature of light emitted by the LEDs of the first light source **301** and the third light source **303** is in the range of approximately 3,000 K to 6,000 K, preferably in the range of approximately 3,500K to 5,500K, more preferably in the range of approximately 4,000K to 5,00, and even more preferably approximately 4,500 K, respectively. Embodiments, however, are not limited thereto, and thus, this disclosure contemplates the light emitted by the LEDs may have any suitable color temperature that falls within the spirit and scope of the principles of this disclosure.

(24) The rechargeable power source **400** is arranged at the intermediate housing region **203**, and comprises a rechargeable battery **401** that is mounted on a PCB **402** and covered by a battery cover member **211**. The rechargeable battery **401** may be one or more 4-volt lithium-based battery cells having a capacity of at least 3 amp.Math.hours. The battery cells are illustratively cylindrical, though other forms of battery cell such as pouch cells may be alternatively utilized.

(25) A power charging port **304** covered by a power charge port cover **212** is located on an outer sidewall surface of the intermediate housing region **203** to facilitate recharging of the rechargeable power source **400**. The power charging port **304** comprises a USB-C port that facilitates a wired connection with a male plug of a USB-C charging cable for recharging of the rechargeable power source **400**. The USB-C port is operable to facilitate a charge rate of 15 watts and a total charging time of 60 minutes to reach a full charged state of the rechargeable power source **400/401**.

(26) A charge indicator **306** is located on the outer sidewall surface of the intermediate housing region **203** to visually indicate a charge status of the rechargeable power source **400** during a recharging sequence. The charge indicator **305** is operable to illuminate a single color that corresponds to the charge status of the rechargeable power source **305**. Alternatively, the charge indicator **305** is operable to illuminate a plurality of colors that correspond to the charge status of the rechargeable power source **400**.

(27) The portable flashlight apparatus **100** includes a user interface (UI) comprising a first control switch member **306** located on an outer sidewall surface of the front housing region **201** to control activation/deactivation of the LEDs of the first light source **301** and the third light source **303**, respectively between a plurality of operating modes. The one or more LEDs of the first light source **301** and the third light source **303** are operable for manual control by a user via the first control switch member **306** to emit illuminating light. The first control switch member **306** is also operable to control an output/intensity of light emitted by the first light source **301** and the third light source **303** between the plurality of operating modes.

(28) For example, in a first or maximum (high) spotlight operating mode of the first light source **301**, an output of the first light source **301** is at least approximately 100 lumens, preferably at least 1200 lumens, with a runtime of at least approximately 2 hours, preferably at least approximately 4 hours. In a second or minimum (low) spotlight operating mode, an output of the first light source **301** is at least approximately 150 lumens, preferably at least 200 lumens, with a runtime of at least approximately 6 hours, preferably at least approximately 8 hours. In a third or floodlight operating mode, an output of the third light source **303** is at least approximately 300 lumens, preferably at least approximately 400 lumens, with a runtime of at least approximately 4 hours, preferably at least approximately 6 hours. In a fourth or hybrid operating mode, the light output is at least approximately 300 lumens, preferably at least approximately 400 lumens, with a runtime of at least approximately 3 hours, preferably at least approximately hours. In at least the first operating mode, the light from the third light source **303** is capable of reaching a beam distance of at least approximately 150 meters, preferably at least approximately 175 meters.

(29) The UI also includes a second control switch member **307** arranged adjacent to the first control switch member **306** on the outer sidewall surface of the front housing region **201** to control activation/deactivation of the laser light of the second light source **302** between the plurality of operating modes.

ADDITIONAL NOTES AND EXAMPLES

(30) Example 1 may include a portable flashlight apparatus, comprising: a base housing defining an interior space, the base housing including structurally defined regions that include a front housing region, a rear housing region, an intermediate housing region extending between the front housing region and the rear housing region and an end cap that closes the interior space at the rear housing region; a light module arranged at the front housing region and operable between a plurality of operating modes, the light module including a first light source operable to emit illuminating light from the front housing region in a direction parallel to the longitudinal axis of the base housing, a second light source operable to emit laser light from the front housing region in a direction parallel to the longitudinal axis of the base housing, and a third light source operable to emit illuminating light from the front housing region in a direction perpendicular to the longitudinal axis of the base housing; a clip member extending longitudinally along outer sidewall surfaces of the rear housing region and the intermediate housing region to facilitate secure attachment of the portable flashlight apparatus to a user; a hook member, arranged on a rear face of the end cap and pivotably moveable between a stowed position on the rear face of the end cap and a deployed position in a direction away from the rear face of the end cap in order to facilitate hanging or suspension of the portable flashlight apparatus at a support point; and one or more magnetic elements, arranged at the rear face of the end cap, and operable to magnetically bias the base housing on a working surface or support surface having magnetic properties. Example 2 may include the portable flashlight

apparatus of Example 1, wherein the front housing region has a first outer perimeter, the rear housing region has a second outer perimeter, and the intermediate housing region has a third outer perimeter that is less than the first outer perimeter and the second outer perimeter. Example 3 may include the portable flashlight apparatus of Example 2, wherein the third outer perimeter of the intermediate housing region defines a gripping surface that permits a user to maintain a grip of the base housing. Example 4 may include the portable flashlight apparatus of Example 1, wherein the first light source comprises one or more LEDs. Example 5 may include the portable flashlight apparatus of Example 4, wherein the color temperature of light emitted by the LEDs of the first light source is 4,500 K. Example 6 may include the portable flashlight apparatus of Example 1, wherein the second light source comprises a laser. Example 7 may include the portable flashlight apparatus of Example 1, wherein the third light source comprises one or more LEDs. Example 8 may include the portable flashlight apparatus of Example 7, wherein the color temperature of light emitted by the LEDs of the third light source is 4,500 K. Example 9 may include the portable flashlight apparatus of Example 1, further comprising a rechargeable power source arranged at the intermediate housing region. Example 10 may include the portable flashlight apparatus of Example 9, wherein the rechargeable power source comprises a rechargeable battery. Example 11 may include the portable flashlight apparatus of Example 9, further comprising a power charging port arranged on an outer sidewall surface of the intermediate housing region to facilitate recharging of the rechargeable battery. Example 12 may include the portable flashlight apparatus of Example 11, wherein the power charging port comprises a USB-C port operable to facilitate a total charging time of 60 minutes to reach a full charged state of the rechargeable battery. Example 13 may include the portable flashlight apparatus of Example 11, further comprising a charge indicator arranged on the outer sidewall surface of the intermediate housing region to visually indicate a charge status of the rechargeable power source during a recharging sequence. Example 14 may include the portable flashlight apparatus of Example 13, wherein the charge indicator is operable to illuminate a single color that corresponds to a charge status of the rechargeable power source. Example 15 may include the portable flashlight apparatus of Example 13, wherein the charge indicator is operable to illuminate a plurality of colors that correspond to a charge status of the rechargeable power source. Example 16 may include the portable flashlight apparatus of Example 1, further comprising a user interface (UI) arranged on an outer sidewall surface of the front housing region. Example 17 may include the portable flashlight apparatus of Example 16, wherein the user interface comprises: a first control switch member operable to control activation/deactivation of the LEDs of the first light source and the third light source, respectively between the plurality of operating modes, and a second control switch member arranged adjacent to the first control switch member on the outer sidewall surface of the front housing region and operable to control activation/deactivation of the laser light of the second light source between the plurality of operating modes. Example 18 may include the portable flashlight apparatus of Example 17, wherein the first control switch member is operable to control an output/intensity of light emitted by the first light source and the third light source between the plurality of operating modes. Example 1B may include a portable flashlight apparatus, comprising: a base housing defining an interior space, the base housing including structurally defined regions that include a front housing region, a rear housing region, an intermediate housing region extending between the front housing region and the rear housing region and an end cap that closes the interior space at the rear housing region; a light module arranged at the front housing region and operable between a plurality of operating modes, the light module including a first light source operable to emit illuminating light from the front housing region in a direction parallel to the longitudinal axis of the base housing, a second light source operable to emit laser light from the front housing region in a direction parallel to the longitudinal axis of the base housing, and a third light source operable to emit illuminating light from the front housing region in a direction perpendicular to the longitudinal axis of the base housing; and one or more magnetic elements, arranged at the rear face of the end cap, and operable

to magnetically bias the base housing on a working surface or support surface having magnetic properties. Example 2B may include the portable flashlight apparatus of Example 1 B, wherein the front housing region has a first outer perimeter, the rear housing region has a second outer perimeter, and the intermediate housing region has a third outer perimeter that is less than the first outer perimeter and the second outer perimeter. Example 3B may include the portable flashlight apparatus of Example 2B, wherein the third outer perimeter of the intermediate housing region defines a gripping surface that permits a user to maintain a grip of the base housing. Example 4B may include the portable flashlight apparatus of Example 1 B, wherein the first light source comprises one or more LEDs. Example 5B may include the portable flashlight apparatus of Example 4B, wherein the color temperature of light emitted by the LEDs of the first light source is 4,500 K. Example 6B may include the portable flashlight apparatus of Example 1 B, wherein the second light source comprises a laser. Example 7B may include the portable flashlight apparatus of Example 1 B, wherein the third light source comprises one or more LEDs. Example 8B may include the portable flashlight apparatus of Example 7B, wherein the color temperature of light emitted by the LEDs of the third light source is 4,500 K. Example 9B may include the portable flashlight apparatus of Example 1 B, further comprising a rechargeable power source arranged at the intermediate housing region. Example 10B may include the portable flashlight apparatus of Example 9B, wherein the rechargeable power source comprises a rechargeable battery. Example 11B may include the portable flashlight apparatus of Example 9B, further comprising a power charging port arranged on an outer sidewall surface of the intermediate housing region to facilitate recharging of the rechargeable battery. Example 12B may include the portable flashlight apparatus of Example 11 B, wherein the power charging port comprises a USB-C port operable to facilitate a total charging time of 60 minutes to reach a full charged state of the rechargeable battery. Example 13B may include the portable flashlight apparatus of Example 11 B, further comprising a charge indicator arranged on the outer sidewall surface of the intermediate housing region to visually indicate a charge status of the rechargeable power source during a recharging sequence. Example 14B may include the portable flashlight apparatus of Example 13B, wherein the charge indicator is operable to illuminate a single color that corresponds to a charge status of the rechargeable power source. Example 15B may include the portable flashlight apparatus of Example 13B, wherein the charge indicator is operable to illuminate a plurality of colors that correspond to a charge status of the rechargeable power source. Example 16B may include the portable flashlight apparatus of Example 1 B, further comprising a user interface (UI) arranged on an outer sidewall surface of the front housing region. Example 17B may include the portable flashlight apparatus of Example 16B, wherein the user interface comprises: a first control switch member operable to control activation/deactivation of the LEDs of the first light source and the third light source, respectively between the plurality of operating modes, and a second control switch member arranged adjacent to the first control switch member on the outer sidewall surface of the front housing region and operable to control activation/deactivation of the laser light of the second light source between the plurality of operating modes. Example 18B may include the portable flashlight apparatus of Example 17B, wherein the first control switch member is operable to control an output/intensity of light emitted by the first light source and the third light source between the plurality of operating modes. Example 1C may include a portable flashlight apparatus, comprising: a housing defining an interior space; a light module arranged at a front end of the housing, a rechargeable battery cell disposed within the interior space, and a charging port arranged to receive a charging cable for charging the battery cell, wherein the light module is configured to output at least approximately 1,000 lumens for at least approximately 2 hours from the battery cell, and wherein the total weight of the flashlight apparatus is less than or equal to approximately 0.62 lbs. Example 2C may include the portable flashlight apparatus of Example 1C, wherein a beam of light from the light module reaches a distance of at least 150 meters. Example 3C may include the portable flashlight apparatus of Example 1C, wherein the charging port is a USB-C port operable to facilitate a charge rate of 15

watts and a total charging time of 60 minutes to reach a full charged state of the rechargeable battery cell.

(31) The terms “coupled,” “attached,” or “connected” can be used herein to refer to any type of relationship, direct or indirect, between the components in question, and can apply to electrical, mechanical, fluid, optical, electromagnetic, electromechanical, or other connections. Additionally, the terms “first,” “second,” etc. are used herein only to facilitate discussion, and carry no particular temporal or chronological significance unless otherwise indicated. The terms “cause” or “causing” means to make, force, compel, direct, command, instruct, and/or enable an event or action to occur or at least be in a state where such event or action can occur, either in a direct or indirect manner.

(32) Those skilled in the art will appreciate from the foregoing description that the broad techniques of the embodiments of the present disclosure can be implemented in a variety of forms. Therefore, while the embodiments of this disclosure have been described in connection with particular examples thereof, the true scope of the embodiments of the disclosure should not be so limited since other modifications will become apparent to the skilled practitioner upon a study of the drawings and specification.

(33) TABLE-US-00001 ITEM PERFORMANCE CHARACTERISTICS Power source 18650 Cell 4 V 3000 mAh Brightness Max. Spot High: 1,200 lm Min. Spot High: 1000 lm Max. Spot Low: 200 lm Min. Spot Low: 150 lm Max. Flood: 250 lm Min. Flood: 150 lm Runtime Max. Spot High: 4 hrs Min. Spot High: 2 hrs Max. Spot Low: 8 hrs Min. Spot Low: 6 hrs Max. Flood: 6 hrs Min. Flood: 4 hrs Charge time 67 min = 100% (Min. -100min) Weight <0.62 lbs Power Source Cycle Life 400 cycles

Claims

1. A portable flashlight apparatus, comprising: a base housing defining an interior space, the base housing including structurally defined regions that include a front housing region, a rear housing region, an intermediate housing region extending between the front housing region and the rear housing region and an end cap that closes the interior space at the rear housing region; a light module arranged at the front housing region and operable between a plurality of operating modes, the light module including a first light source operable to emit illuminating light from the front housing region in a direction parallel to a longitudinal axis of the base housing, and a second light source operable to emit illuminating light from the front housing region in a direction perpendicular to the longitudinal axis of the base housing; a magnetic element arranged at the rear face of the end cap, and operable to magnetically bias the base housing on a working surface or support surface having magnetic properties; and a holder in the interior space defined by the base housing; wherein the light module is mounted on the holder; wherein the first light source faces a first direction; wherein the second light source faces a second direction, perpendicular to the first direction; wherein the front housing region has a first outer perimeter, the rear housing region has a second outer perimeter, and the intermediate housing region has a third outer perimeter that is less than the first outer perimeter and the second outer perimeter; and a power charging port arranged centrally on an outer sidewall surface of the intermediate housing region facing in the second direction.
2. The portable flashlight apparatus of claim 1, wherein the first light source comprises one or more LEDs.
3. The portable flashlight apparatus of claim 1, wherein the second light source comprises a plurality of LEDs.
4. The portable flashlight apparatus of claim 3, wherein the color temperature of light emitted by the LEDs of the second light source is 4,500 K.
5. The portable flashlight apparatus of claim 1, further comprising a rechargeable power source arranged at the intermediate housing region; wherein the rechargeable power source is mounted on a printed circuit board.

6. The portable flashlight apparatus of claim 5, wherein the rechargeable power source comprises a rechargeable battery, and wherein the printed circuit board extends along an axis parallel to the longitudinal axis of the base housing; and wherein the printed circuit board is longer than the rechargeable battery.
7. The portable flashlight apparatus of claim 6, wherein the power charging port is operable to facilitate a total charging time of 60 minutes to reach a full charged state of the rechargeable battery.
8. The portable flashlight apparatus of claim 1, further comprising a user interface (UI) arranged on an outer sidewall surface of the front housing region.
9. The portable flashlight apparatus of claim 1, further comprising a hook; wherein when the hook is in the stowed position, the end cap projects farther rearwardly than the hook.
10. The portable flashlight apparatus of claim 1, further comprising a clip member extending longitudinally along an outer sidewall surface of the rear housing region and an outer sidewall surface of the intermediate housing region.
11. A portable flashlight apparatus, comprising: a base housing defining an interior space, the base housing including structurally defined regions that include a front housing region, a rear housing region, an intermediate housing region extending between the front housing region and the rear housing region and an end cap that closes the interior space at the rear housing region; a light module arranged at the front housing region and operable between a plurality of operating modes, the light module including a first light source operable to emit illuminating light from the front housing region in a direction parallel to a longitudinal axis of the base housing, and a second light source operable to emit illuminating light of at least approximately 300 lumens from the front housing region in a direction perpendicular to the longitudinal axis of the base housing; and one or more magnetic elements, arranged at the rear face of the end cap, and operable to magnetically bias the base housing on a working surface or support surface having magnetic properties; wherein the first light source comprises a first LED on a first printed circuit board; wherein the second light source comprises a second LED on a second printed circuit board; wherein the first printed circuit board is spaced apart from the second printed circuit board; wherein the first printed circuit board is on a different plane than the second printed circuit board; further comprising a third printed circuit board; a rechargeable battery on the third printed circuit board; a first seal member arranged between the rear housing region and the end cap; and a second seal member arranged at the front housing region; and a power charging port arranged centrally on an outer sidewall surface of the intermediate housing region facing in the direction of the illuminating light of the second light source.
12. The portable flashlight apparatus of claim 11, wherein the first light source comprises one or more LEDs.
13. The portable flashlight apparatus of claim 11, wherein the second light source comprises one or more LEDs.
14. The portable flashlight apparatus of claim 11, further comprising a user interface (UI) arranged on an outer sidewall surface of the front housing region.
15. The portable flashlight of claim 11, wherein the base housing has a generally rectangular cross-section.
16. The portable flashlight apparatus of claim 11, wherein the front housing region has a first outer perimeter, the rear housing region has a second outer perimeter, and the intermediate housing region has a third outer perimeter that is less than the first outer perimeter and the second outer perimeter; wherein the second outer perimeter is forward of the end cap.
17. A portable flashlight apparatus, comprising: a housing defining an interior space; a light module arranged at a front end of the housing, a rechargeable battery cell disposed within the interior space, and a charging port arranged to receive a charging cable for charging the battery cell, wherein the light module includes a first light source facing in a first direction and a second light

source facing in a second direction perpendicular to the first direction, wherein the first light source is configured to output at least approximately 1,000 lumens for at least approximately 2 hours from the battery cell, and wherein the total weight of the flashlight apparatus is less than or equal to approximately 0.62 lbs; wherein the housing comprises structurally defined regions including a front housing region, a rear housing region, an intermediate housing region extending between the front housing region and the rear housing region and an end cap that closes the interior space at the rear housing region; wherein the front housing region has a first outer perimeter, the rear housing region has a second outer perimeter, and the intermediate housing region has a third outer perimeter that is less than the first outer perimeter and the second outer perimeter; wherein the second outer perimeter is forward of the end cap; and a power charging port arranged centrally on an outer sidewall surface of the intermediate housing region facing in the second direction.

18. The portable flashlight apparatus of claim 17, wherein a beam of light from the light module reaches a distance of at least 150 meters.

19. The portable flashlight apparatus of claim 17, wherein the charging port is operable to facilitate a charge rate of 15 watts and a total charging time of 60 minutes to reach a full charged state of the rechargeable battery cell.

20. The portable flashlight of claim 17, further comprising a holder; wherein the light module is mounted on the holder; wherein the rechargeable battery cell is mounted on the holder.
